

1. Openvas_log

Timestamp	Target IP	Vulnerability	PTES Phase
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2026-04-24 12:00:00	192.168.31.199	XSS	Exploitation
2026-04-24 12:05:00	192.168.31.199	SQL Injection	Exploitation
2026-04-24 12:10:00	192.168.31.199	Outdated Apache	Scanning

2. Ptes_report

A full Vulnerability Assessment and Penetration Testing (VAPT) was conducted on the target system 192.168.31.199 following the PTES methodology. During the reconnaissance phase, information about the target environment and exposed services was gathered using tools such as Shodan and Maltego. The scanning phase identified multiple open ports and services including Apache, Tomcat, and SSH.

In the exploitation phase, a vulnerability in the Apache Tomcat Manager application was successfully exploited using Metasploit, resulting in remote access to the system via a Meterpreter session. This demonstrated the impact of weak credentials and exposed services.

Post-exploitation activities included navigating the system and collecting sensitive files such as /etc/passwd. The file was downloaded and hashed to ensure integrity. Privilege escalation attempts were made but were unsuccessful due to restricted permissions.

SQL injection testing was attempted using sqlmap on the DVWA application. However, due to session instability and environmental limitations, the injection could not be fully confirmed.

Overall, the assessment identified critical security issues that require immediate remediation.

3. Non_technical_summary

A security assessment was conducted on the target system to identify potential vulnerabilities. The testing revealed that the system contains weaknesses such as outdated software and insecure configurations. These issues allowed controlled access to the system during testing.

Although some protections limited deeper access, the system remains at risk of exploitation. Sensitive information could potentially be exposed if these vulnerabilities are not addressed.

To improve security, it is recommended to update all software, enforce stronger authentication mechanisms, and apply proper input validation. Addressing these issues will significantly reduce the risk of cyber threats and unauthorized access.

4. Workflow

1. Performed reconnaissance using Maltego and Shodan
2. Identified target services and IP address
3. Conducted vulnerability scanning using Nmap and Nikto
4. Identified Apache and Tomcat services
5. Exploited Tomcat using Metasploit
6. Gained Meterpreter session
7. Performed post-exploitation and collected /etc/passwd
8. Generated hash for integrity verification
9. Attempted SQL injection using sqlmap
10. Documented all findings and results